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SN123: MANTIS, collective intelligence for financial forecasting

A first look at MANTIS, Bittensor's incentivized 1-hour return forecaster, with time-lock encryption, salience-weighted rewards, and a roadmap to forex and beyond.

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financial forecasting A first look at MANTIS, Bittensor's incentivized 1-hour return forecaster, with time-lock encryption, salience-weighted rewards, and a roadmap to forex and beyond. 2025-08-30 · filed by Subnet Magazine

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At its core, MANTIS is a complex, highly-technical subnet, bringing together the collective intelligence and refined models of hundreds of miners to cultivate a robust financial market prediction tool. At first glance, the subnet seems almost incomprehensible to the average retail investor, but the subnet's true value, its capability, and its strong use- case is seen only upon closer inspection.

MANTIS was founded by MBarbarian7676, a young developer based out of Costa Rica, who had been mining MEthereum, and later, MBittensor in the early days of the blockchain. His experience led him to joining SN8 before finally launching his own subnet. MANTIS began its journey, registering only in June of this year, working to forecast BTC/USD and increased its purview to include multiple assets shortly after.

The MANTIS system aims to forecast only the next 1-hour of returns of

At the start of a time-block, MANTIS accepts miners' predictions for what the price change for a specific asset will be an hour from now. MANTIS then encrypts the miners' data and locks it.

Barbarian MBarbarian7676 · Aug 2, 2025 Currently being built out internally:

Dashboard for token holders to allocate emissions

Switch from Drand to a private timelock server, so that we can access signals the moment they are being generated

A much more scalable build, ultimately allowing for many more challenges of

Punisher ▣▣ MCryptoZPunisher · Aug 2, 2025

On my radar: #SN123 Mantis It's all a matter of patience... A real banger is coming.

MMANTeSBittensor

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An interesting facet of the MANTIS workflow that has now been updated from Drand is the "time-lock encryption" feature that prevents miners from seeing actual price outcomes before their predictions are locked, and only decrypts their submissions when it's time to evaluate them. After the decryption stage, the subnet will train an AI model to predict returns (on a 1-hour basis) using the combined data of all miners. Once the data is used to predict the return, they use a machine-learning (ML) concept called "permutation importance" to measure by how much one miner's data contributed to the overall accuracy. Those deemed to

have contributed the most are deemed to have high "salience". MANTIS rewards those miners, linking their share of the reward (weight setting) to how useful their prediction was. Technical details can be found readily on the subnet's GitHub.

MANTIS' design includes multiple protections against those trying to work against the overall mission. MANTIS verifies that each payload comes from an original miner (tracked through their hotkey), and rejects outliers, and filters out miners who mislead with "garbage data".

MANTIS' miners are incentivized to improve their models over time, to provide the best, most accurate predictions. Every miner acts as its own small-time trader, with its own algorithms, and MANTIS combines them all, getting better and better predictions over time. MANTIS workflow is a coming-together of many working units, multiple components that come together to deliver its final product.

1. validator.py is its main process, that controls the scheduling and execution of the entire process
2. cycle.py and comms.py are in charge of the "ingestion" of the miners' predictions
3. storage.py - this component is responsible for encryption and decryption of the miner's predictions.
4. model.py is MANTIS' AI trainer, based off of XGBoost, that takes the predictions, learns from them, and predicts the return based on all miners' inputs. That model then uses the "permutation importance" to see how "salient" (useful) each miner's input was. It then assigns each miner a "score"
5. Blockchain Interface - this is the final stage, where MANTIS takes the "salience" scores and provides incentives based on those scores

There are a few technical features that showcase the thought that was put into the design of this subnet. The MANTIS architecture is connected directly to Bittensor's block time. The "modular" nature of the programming means the team can adapt or change one piece of the workflow without disrupting the other parts. This signals that the team will be looking at making changes to the workflow over time.

MANTIS also hints at expanding its offerings to more than just MBitcoin, bringing in financial modeling for other cryptocurrencies, and even multiple forex pairs. MANTIS' model has worked well on forex, providing above average accuracy with their current model.

Last month, following an in-depth feature on MOpentensor Foundation's Novelty Search, Barbarian announced that the subnet will be introducing novel volatility modeling tasks for miners, with the revenue from the exercises set to be burnt to support the price of the subnet's alpha. Holding the alpha would also mean being able to be a part of their governance system, getting voting rights on which assets to go after next, or which model to use for aggregating predictions.

Since then, the subnet has strengthened its decentralization and verifiability in the protocol, with signals now double-encrypted with HPKE and a public timelock. The new methods will allow the Subnet owner to instantly decrypt miner payloads in V2. These V2 payloads are now being enforced on the MANTIS validator. Miners were given a deadline of October 20 to migrate to the V2 version. Once approximately 90% of the miners have migrated, the auction will launch and revenues will follow.

The auction feature's frontend is being polished, with MBarbarian7676 sharing that there "are no blocks on the network preventing us from starting the auctions" and the next step being launch. According to a post from October 13th, the performance data for the subnet were 0.6 AUC (At-auction Order) on some assets, with an average above 0.55 AUC.

MANTIS' goal is to have revenue from the auction go into buybacks of the alpha token, and inevitable burn to support the price of the token.

Finally, after much anticipation, on October 24, MBarbarian7676 released the details of the miners' new prediction target:

MANTIS is exciting, in that it offers a definitively monetizable use case for financial institutions, hedge funds, traders, that want highly-accurate prediction models for prices in the market. Private models range from tens of thousands for simpler custom solutions to millions for large, complex models. MANTIS is doing this with the power of open source tech, for a small fractional cost. The subnet since its launch less than four months ago, has shown steady growth, reflected in the price of its alpha token.

When, not if, the model is refined and packaged for market, it could make waves, disrupting the closed-box, private institutions that charge top-dollar for their prediction models. While this is not the most user-forward subnet, with its highly technical language, MANTIS posits itself as a front-runner amongst subnets that could bring real value to clients that come with big money. If MANTIS' draw is their predictive financial models, it won't be long before large institutions with deep pockets look into the ecosystem it was built out of.

MBarbarian7676 has also just hinted at his acquisition of SN125 and his new project, 8Ball - the "first prediction market powered by MBittensor". The website and markets will be launched soon, including predictive markets for sports, politics, crypto, as well as a section for societal issues called "1984" focused on censorship, freedom of speech, and state power. Punisher · MCryptoZPunisher · Oct 15, 2025 Bittensor \$TAO

Some news from MBarbarian7676 , SN125 8 Ball is coming.

Here's a summary.

8 Ball is preparing to launch its website and its first markets, likely within the next 3 days.

The project will combine predictive markets focused on politics, sports, crypto, as

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This signals that the story for MBarbarian7676 and his subnets is just beginning, and we're excited to see what's next for the young developer.

Article submitted by: @subnetmagazine team